

Stats 596 – Homework 2

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Due Thursday, October 8, 2026

1. Exercise 1.11.

Let $Z \sim N(\mu, 1)$, where $\mu \in \mathbb{R}$.

(a) For $\lambda \geq 0$, define the ridge estimator

$$\hat{\mu}_\lambda(z) = \operatorname{argmin}_{t \in \mathbb{R}} \left\{ \frac{1}{2}(t - z)^2 + \frac{\lambda}{2}t^2 \right\}.$$

Compute

$$\operatorname{MSE}_\lambda(\mu) = \mathbb{E}\{(\hat{\mu}_\lambda(Z) - \mu)^2\}.$$

Plot and compare the risk functions for $\lambda = 0.1, 1, 5, 10$.

(b) Repeat for the Lasso estimator

$$\hat{\mu}_\lambda(z) = \operatorname{argmin}_{t \in \mathbb{R}} \left\{ \frac{1}{2}(t - z)^2 + \lambda|t| \right\}.$$

Solution:

(a) **Ridge.** The derivative of the objective is

$$(t - z) + \lambda t = (1 + \lambda)t - z.$$

Its second derivative is $1 + \lambda > 0$, so the unique minimizer is

$$\boxed{\hat{\mu}_\lambda(z) = \frac{z}{1 + \lambda}}.$$

Since $\mathbb{E}Z = \mu$ and $\operatorname{Var}(Z) = 1$,

$$\mathbb{E}\hat{\mu}_\lambda(Z) - \mu = -\frac{\lambda\mu}{1 + \lambda}, \quad \operatorname{Var}\{\hat{\mu}_\lambda(Z)\} = \frac{1}{(1 + \lambda)^2}.$$

Therefore

$$\text{MSE}_\lambda^{\text{ridge}}(\mu) = \frac{1 + \lambda^2 \mu^2}{(1 + \lambda)^2}.$$

Each risk curve is even in μ and has its minimum at zero. Larger penalties reduce variance near zero but increase shrinkage bias for sufficiently large signals.

The unpenalized estimator Z has MSE 1. For $\lambda > 0$,

$$\text{MSE}_\lambda^{\text{ridge}}(\mu) < 1 \iff 1 + \lambda^2 \mu^2 < (1 + \lambda)^2 \iff \mu^2 < 1 + \frac{2}{\lambda}.$$

Thus a positive ridge penalty improves on Z within a bounded range of signal sizes.

At $\mu = 0$, the risks for the four penalties are approximately

$$0.826446, \quad 0.25, \quad 0.027778, \quad 0.008264.$$

(b) Lasso. The stationary condition is

$$0 \in t - z + \lambda \partial |t|.$$

Considering positive, zero, and negative t gives

$$\hat{\mu}_\lambda(z) = S_\lambda(z) = \begin{cases} z - \lambda, & z > \lambda, \\ 0, & |z| \leq \lambda, \\ z + \lambda, & z < -\lambda. \end{cases}$$

Write $Z = \mu + U$, where $U \sim N(0, 1)$, and define

$$a = -\lambda - \mu, \quad b = \lambda - \mu.$$

Then

$$\hat{\mu}_\lambda(Z) - \mu = \begin{cases} U + \lambda, & U < a, \\ -\mu, & a \leq U \leq b, \\ U - \lambda, & U > b. \end{cases}$$

Let ϕ and Φ denote the standard normal density and CDF. The MSE is

$$\begin{aligned} \text{MSE}_\lambda^{\text{lasso}}(\mu) &= \int_{-\infty}^a (u + \lambda)^2 \phi(u) du \\ &\quad + \mu^2 \{\Phi(b) - \Phi(a)\} \\ &\quad + \int_b^{\infty} (u - \lambda)^2 \phi(u) du. \end{aligned}$$

Using $\phi'(u) = -u\phi(u)$ and integration by parts,

$$\begin{aligned}\int_{-\infty}^a u\phi(u) du &= -\phi(a), \\ \int_{-\infty}^a u^2\phi(u) du &= \Phi(a) - a\phi(a), \\ \int_b^{\infty} u\phi(u) du &= \phi(b), \\ \int_b^{\infty} u^2\phi(u) du &= 1 - \Phi(b) + b\phi(b).\end{aligned}$$

Expanding the squares and substituting yields

$$\begin{aligned}\text{MSE}_{\lambda}^{\text{lasso}}(\mu) &= (1 + \lambda^2)\{\Phi(a) + 1 - \Phi(b)\} \\ &\quad - (a + 2\lambda)\phi(a) \\ &\quad + (b - 2\lambda)\phi(b) \\ &\quad + \mu^2\{\Phi(b) - \Phi(a)\}.\end{aligned}$$

At $\lambda = 0$, the formula reduces to 1. At $\mu = 0$, it becomes

$$\text{MSE}_{\lambda}^{\text{lasso}}(0) = 2\{(1 + \lambda^2)\Phi(-\lambda) - \lambda\phi(\lambda)\}.$$

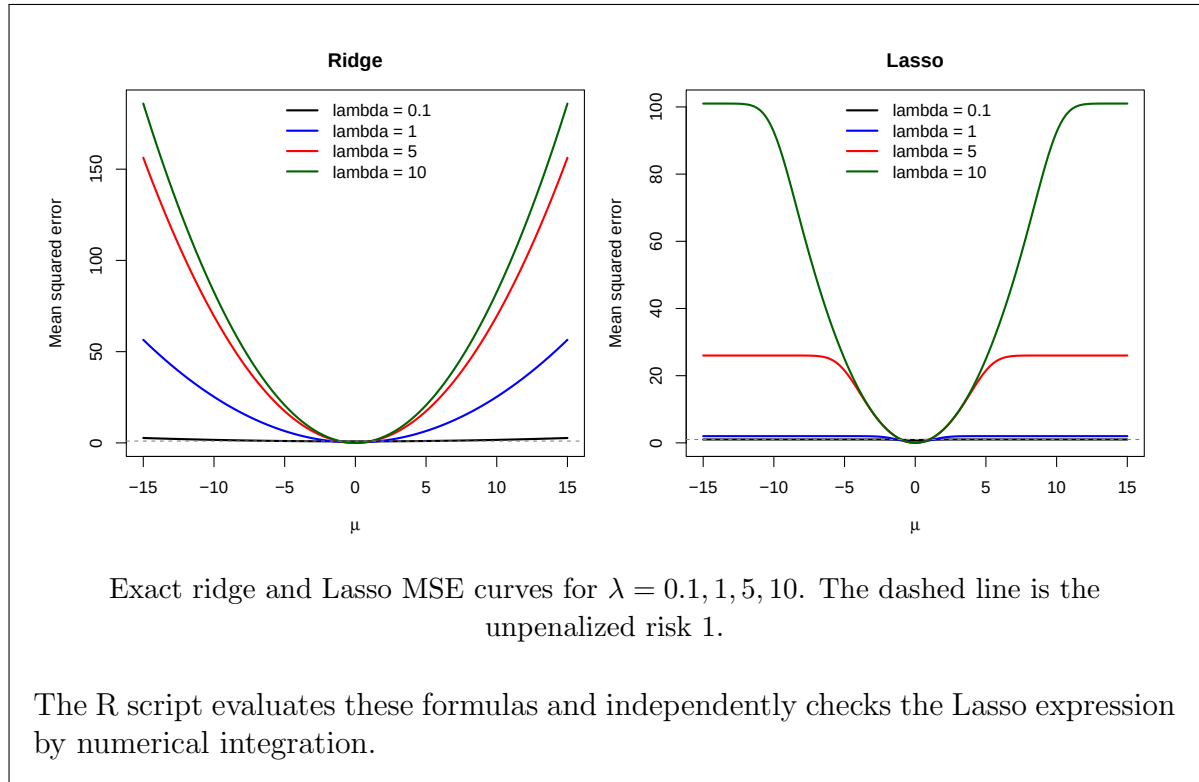
The risk is even in μ . Large penalties produce small risk near zero by setting the estimator to zero with high probability. For fixed λ ,

$$\lim_{|\mu| \rightarrow \infty} \text{MSE}_{\lambda}^{\text{lasso}}(\mu) = 1 + \lambda^2.$$

The limiting risks for the four penalties are

$$1.01, \quad 2, \quad 26, \quad 101.$$

Unlike ridge, the Lasso risk approaches a finite plateau for each fixed penalty.



2. Exercise 1.12.

Let Y_i be responses and let $X_i = X_{i,1:p}$ denote covariates excluding a constant. Define

$$\tilde{Y}_i = Y_i - \bar{Y}, \quad \tilde{X}_i = X_i - \bar{X}.$$

- (a) For ridge regression with an unpenalized intercept, show that the slopes can be obtained from the centered data and that

$$\hat{\beta}_0 = \bar{Y} - \bar{X}^\top \tilde{\beta}_{1:p}.$$

Also show that

$$\text{tr}(S_\lambda) = 1 + \text{tr}(\tilde{S}_\lambda).$$

- (b) Prove the same centering and intercept identities for Lasso.

Solution:

(a) Write $b = \beta_{1:p}$. The ridge criterion is

$$Q(\beta_0, b) = \frac{1}{2n} \sum_{i=1}^n (Y_i - \beta_0 - X_i^\top b)^2 + \frac{\lambda}{2} \|b\|_2^2.$$

For fixed b , the penalty is independent of the intercept. Differentiating in β_0 gives

$$\frac{\partial Q}{\partial \beta_0} = -\frac{1}{n} \sum_{i=1}^n (Y_i - \beta_0 - X_i^\top b).$$

The second derivative is 1, so the unique minimizing intercept is

$$\beta_0(b) = \bar{Y} - \bar{X}^\top b.$$

Substituting into the residual gives

$$Y_i - \beta_0(b) - X_i^\top b = \tilde{Y}_i - \tilde{X}_i^\top b.$$

Consequently the profiled objective is

$$\min_{\beta_0} Q(\beta_0, b) = \frac{1}{2n} \sum_{i=1}^n (\tilde{Y}_i - \tilde{X}_i^\top b)^2 + \frac{\lambda}{2} \|b\|_2^2.$$

This is exactly the centered ridge problem. Therefore

$$\boxed{\hat{\beta}_{1:p} = \tilde{\beta}_{1:p}, \quad \hat{\beta}_0 = \bar{Y} - \bar{X}^\top \tilde{\beta}_{1:p}.}$$

Any centered minimizer gives a joint minimizer, even when the slope solution is not unique at $\lambda = 0$.

For the smoother identity, let X_c be the original $n \times p$ covariate matrix and define

$$J = \frac{1}{n} \mathbf{1}\mathbf{1}^\top, \quad H = I - J.$$

Then

$$\tilde{X} = HX_c, \quad \tilde{Y} = HY.$$

For $\lambda > 0$, the centered smoother is

$$\tilde{S}_\lambda = \tilde{X}(\tilde{X}^\top \tilde{X} + n\lambda I)^{-1} \tilde{X}^\top.$$

Because $\tilde{X}^\top \mathbf{1} = 0$,

$$\tilde{S}_\lambda J = 0, \quad \tilde{S}_\lambda H = \tilde{S}_\lambda.$$

The fitted response from the original regression is

$$\begin{aligned} \hat{Y} &= \mathbf{1}\hat{\beta}_0 + X_c \tilde{\beta}_{1:p} \\ &= \bar{Y}\mathbf{1} + \tilde{X} \tilde{\beta}_{1:p} \\ &= JY + \tilde{S}_\lambda HY \\ &= (J + \tilde{S}_\lambda)Y. \end{aligned}$$

Thus

$$S_\lambda = J + \tilde{S}_\lambda.$$

Since $\text{tr}(J) = 1$,

$$\text{tr}(S_\lambda) = 1 + \text{tr}(\tilde{S}_\lambda).$$

At $\lambda = 0$, use the centered least-squares projection matrix, defined with a Moore–Penrose inverse when necessary. The same fitted-value argument applies.

(b) For Lasso, the objective becomes

$$Q(\beta_0, b) = \frac{1}{2n} \sum_{i=1}^n (Y_i - \beta_0 - X_i^\top b)^2 + \lambda \|b\|_1.$$

The penalty still does not involve β_0 , so the optimal intercept and residual substitution are unchanged. Hence

$$\hat{\beta}_{1:p} = \tilde{\beta}_{1:p}, \quad \hat{\beta}_0 = \bar{Y} - \bar{X}^\top \tilde{\beta}_{1:p}.$$

No differentiation of the Lasso penalty is needed.

3. Exercise 1.13.

Let

$$s_j = \|X_j\|_n = \left(\frac{1}{n} \sum_{i=1}^n X_{ij}^2 \right)^{1/2}, \quad \tilde{X}_{ij} = \frac{X_{ij}}{s_j}.$$

- (a) Find weights c_j so that fitting an unweighted ridge penalty to the standardized covariates and setting $\hat{\beta}_j = \hat{\gamma}_j/s_j$ is equivalent to fitting a weighted ridge penalty to the original covariates.
- (b) Find the corresponding weights for Lasso.

Solution:

Assume $s_j > 0$ for each j , since otherwise the stated standardization is undefined.

Let

$$D = \text{diag}(s_1, \dots, s_p).$$

Then $\tilde{X} = XD^{-1}$, and the change of variables

$$\gamma = D\beta$$

is a bijection satisfying

$$\tilde{X}\gamma = X\beta.$$

(a) **Ridge.** The standardized objective becomes

$$\begin{aligned} \frac{1}{2n} \|Y - \tilde{X}\gamma\|_2^2 + \frac{\lambda}{2} \sum_{j=1}^p \gamma_j^2 \\ = \frac{1}{2n} \|Y - X\beta\|_2^2 + \frac{\lambda}{2} \sum_{j=1}^p s_j^2 \beta_j^2. \end{aligned}$$

The objectives agree under the bijection, so their minimizers correspond. Therefore

$$c_j = s_j^2 = \frac{1}{n} \sum_{i=1}^n X_{ij}^2.$$

(b) **Lasso.** Similarly,

$$\begin{aligned} \frac{1}{2n} \|Y - \tilde{X}\gamma\|_2^2 + \lambda \sum_{j=1}^p |\gamma_j| \\ = \frac{1}{2n} \|Y - X\beta\|_2^2 + \lambda \sum_{j=1}^p s_j |\beta_j|. \end{aligned}$$

Thus

$$c_j = s_j = \left(\frac{1}{n} \sum_{i=1}^n X_{ij}^2 \right)^{1/2}.$$

Standardization therefore changes the relative penalty weights on the original coefficients. At $\lambda = 0$, the penalty weights are immaterial, but the stated choices remain valid.

4. Exercise 1.15.

(a) For the Lasso objective

$$h(\beta) = \frac{1}{2n} \|Y - X\beta\|_2^2 + \lambda \|\beta\|_1,$$

show that coordinate-wise minimization implies joint minimization.

(b) Generalize to

$$h(\beta) = h_0(\beta) + \sum_{j=1}^p h_{1,j}(\beta_j),$$

where h_0 is convex and everywhere differentiable, and each $h_{1,j}$ is convex but may be nondifferentiable.

- (c) (Optional.) Give a convex counterexample for which coordinate-wise minimization does not imply joint minimization.

Solution:

(a) Let

$$h_0(\beta) = \frac{1}{2n} \|Y - X\beta\|_2^2.$$

Its gradient is

$$\nabla h_0(\beta) = -\frac{1}{n} X^\top (Y - X\beta).$$

Suppose $\tilde{\beta}$ minimizes the objective in every coordinate when the other coordinates are held fixed. The one-dimensional optimality condition for coordinate j is

$$0 \in \frac{\partial h_0}{\partial \beta_j}(\tilde{\beta}) + \lambda \partial |\tilde{\beta}_j|.$$

The absolute-value subdifferential is

$$\partial |t| = \begin{cases} \{1\}, & t > 0, \\ [-1, 1], & t = 0, \\ \{-1\}, & t < 0. \end{cases}$$

For each j , select $s_j \in \partial |\tilde{\beta}_j|$ such that

$$\frac{\partial h_0}{\partial \beta_j}(\tilde{\beta}) + \lambda s_j = 0.$$

Since the L_1 penalty is separable,

$$s = (s_1, \dots, s_p)^\top \in \partial \|\tilde{\beta}\|_1.$$

Combining the coordinate conditions gives

$$0 = \nabla h_0(\tilde{\beta}) + \lambda s \in \partial h(\tilde{\beta}).$$

By the subgradient inequality, for every β ,

$$h(\beta) \geq h(\tilde{\beta}) + 0^\top (\beta - \tilde{\beta}) = h(\tilde{\beta}).$$

Therefore $\tilde{\beta}$ is a global minimizer.

Equivalently, its stationary conditions are

$$\frac{1}{n} X_j^\top (Y - X\tilde{\beta}) \begin{cases} = \lambda, & \tilde{\beta}_j > 0, \\ \in [-\lambda, \lambda], & \tilde{\beta}_j = 0, \\ = -\lambda, & \tilde{\beta}_j < 0. \end{cases}$$

This establishes the optimality of a coordinate-wise minimizing point; it is not by itself a proof of convergence of every coordinate-descent implementation.

(b) Coordinate-wise optimality implies that, for each j , there exists

$$s_j \in \partial h_{1,j}(\tilde{\beta}_j)$$

satisfying

$$s_j = -\frac{\partial h_0}{\partial \beta_j}(\tilde{\beta}).$$

Convexity of h_0 gives

$$h_0(\beta) \geq h_0(\tilde{\beta}) + \nabla h_0(\tilde{\beta})^\top (\beta - \tilde{\beta}).$$

The subgradient inequality for each penalty term gives

$$h_{1,j}(\beta_j) \geq h_{1,j}(\tilde{\beta}_j) + s_j(\beta_j - \tilde{\beta}_j).$$

Adding these inequalities,

$$\begin{aligned} h(\beta) &\geq h(\tilde{\beta}) + \sum_{j=1}^p \left\{ \frac{\partial h_0}{\partial \beta_j}(\tilde{\beta}) + s_j \right\} (\beta_j - \tilde{\beta}_j) \\ &= h(\tilde{\beta}). \end{aligned}$$

Hence $\tilde{\beta}$ is jointly minimizing. Separability allows the individual penalty subgradients to be chosen simultaneously.

(c) Consider

$$h(x, y) = |x - y| + \frac{1}{4}(x + y - 2)^2.$$

The function is convex because it is the sum of the absolute value of a linear function and the square of an affine function.

At $(0, 0)$, fixing $y = 0$ gives

$$h(x, 0) - h(0, 0) = |x| - x + \frac{x^2}{4} \geq 0.$$

Equality holds only at $x = 0$. By symmetry,

$$h(0, y) - h(0, 0) = |y| - y + \frac{y^2}{4} \geq 0,$$

with equality only at $y = 0$.

Thus $(0, 0)$ uniquely minimizes the objective along each coordinate line. However,

$$h(0, 0) = 1, \quad h(1, 1) = 0.$$

Since $h \geq 0$, $(1, 1)$ is a global minimizer and $(0, 0)$ is not.

The nondifferentiable term $|x - y|$ couples the coordinates; it does not satisfy the separability assumption in part (b).

5. Exercise 1.17.

Generate training and test data using the following code.

```
1 library(MASS)
2 set.seed(0)
3
4 n <- 200
5 p <- 20
6 mu <- rep(0,p)
7 msig <- toeplitz(.8^(1:p-1))
8 sig <- .5
9
10 x <- mvrnorm(n, mu, msig)
11 y <- x[,1] +
12   pmax(0, apply(x[,2:4],1,sum))^2 / 3 +
13   rnorm(n, 0, sig)
14
15 cbind(y,x)[1:3, 1:6]
16
17 ntest <- 100
18 xtest <- mvrnorm(ntest, mu, msig)
19 ytest <- xtest[,1] +
20   pmax(0, apply(xtest[,2:4],1,sum))^2 / 3 +
21   rnorm(ntest, 0, sig)
22
23 cbind(ytest,xtest)[1:3, 1:6]
```

Apply all methods without standardizing x or y .

- Fit ridge regression with an unpenalized intercept and produce a coefficient path.
- Plot training, test, 5-fold CV, and 10-fold CV errors against degrees of freedom. Report the minimum test error and test errors at the minimum-CV and one-standard-error choices.
- Fit Lasso with an unpenalized intercept and produce a coefficient path.
- Plot the four errors against the shrinkage fraction and report the corresponding test-error comparisons.
- Summarize the findings in one paragraph.

Solution:

Data and reproducibility. The covariance is

$$\Sigma_{jk} = 0.8^{|j-k|},$$

and the generating conditional mean is

$$m(X) = X_1 + \frac{1}{3}\{\max(0, X_2 + X_3 + X_4)\}^2.$$

The first three training rows, showing the response and the first five covariates, are

Row	y	x_1	x_2	x_3	x_4	x_5
1	5.249158	0.697133	0.987522	1.607691	0.935633	1.342265
2	5.854535	1.648578	0.964429	1.556518	1.263750	1.205795
3	-0.003703	0.097052	-0.665160	-1.122233	-0.102869	0.052841

The corresponding test rows are

Row	y	x_1	x_2	x_3	x_4	x_5
1	-1.007614	-1.028385	-0.950839	-1.216774	-1.325364	-1.157469
2	0.970894	0.659170	0.380144	-0.132891	0.604044	0.561049
3	0.238813	-0.384081	-0.023682	-1.498525	-0.513272	-0.542364

The R script generates both samples before generating CV folds. It then uses a separate seed of 123 for the fold partition. Ridge and Lasso use the same partitions. Neither x nor y is rescaled; centering profiles out the unpenalized intercept.

(a) Ridge. For each penalty,

$$\hat{b}_\lambda = (\tilde{X}^\top \tilde{X} + n\lambda I)^{-1} \tilde{X}^\top \tilde{Y},$$

and

$$\hat{\beta}_{0,\lambda} = \bar{Y} - \bar{X}^\top \hat{b}_\lambda.$$

The implementation uses an SVD rather than explicitly inverting the Gram matrix. If $d_1, \dots, d_r$ are the retained nonzero singular values of $\tilde{X}$, the effective degrees of freedom are

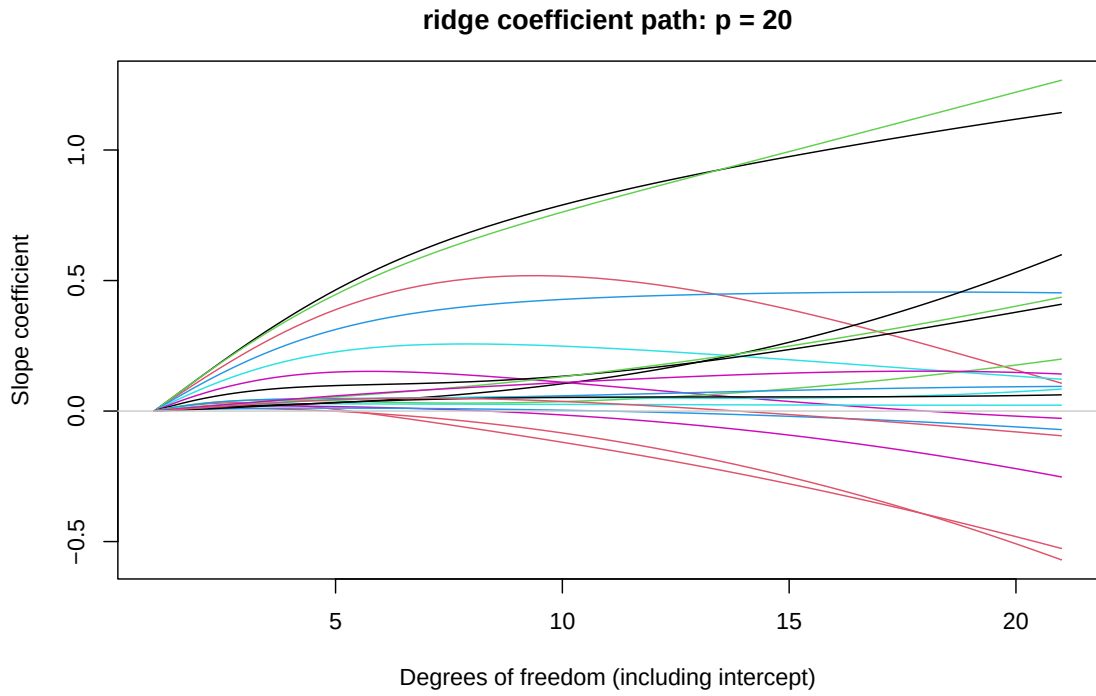
$$\text{df}(\lambda) = 1 + \sum_{j=1}^r \frac{d_j^2}{d_j^2 + n\lambda}.$$

The intercept contributes the leading 1.

The grid contains 240 positive penalties from $10^4 s$ to $10^{-8} s$, where

$$s = \frac{d_{\max}^2}{n}.$$

This sets the penalty-grid scale without standardizing the data.



Ridge coefficient path versus effective degrees of freedom.

(b) Errors and cross-validation. Training and test errors are mean squared prediction errors:

$$\text{Err}_{\text{train}}(\lambda) = \frac{1}{n} \sum_{i=1}^n (Y_i - \hat{Y}_{i,\lambda})^2,$$

$$\text{Err}_{\text{test}}(\lambda) = \frac{1}{n_{\text{test}}} \sum_{i=1}^{n_{\text{test}}} (Y_i^{\text{test}} - \hat{Y}_{i,\lambda}^{\text{test}})^2.$$

For $K \in \{5, 10\}$, let $e_k(\lambda)$ be the mean squared held-out error in fold k , using a fit trained on the other folds. The folds have equal size, so

$$\text{CV}_K(\lambda) = \frac{1}{K} \sum_{k=1}^K e_k(\lambda),$$

and the fold-based standard error is

$$\text{SE}_K(\lambda) = \frac{\text{sd}\{e_1(\lambda), \dots, e_K(\lambda)\}}{\sqrt{K}}.$$

The minimum-CV choice is

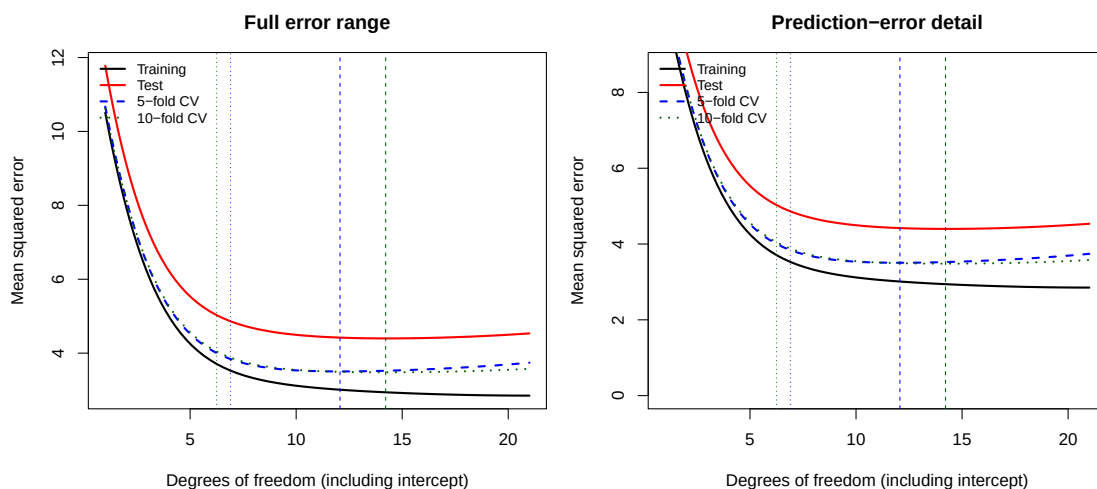
$$\lambda_{\min, K} \in \arg\min_{\lambda \in \Lambda} \text{CV}_K(\lambda).$$

The one-standard-error choice is the largest penalty satisfying

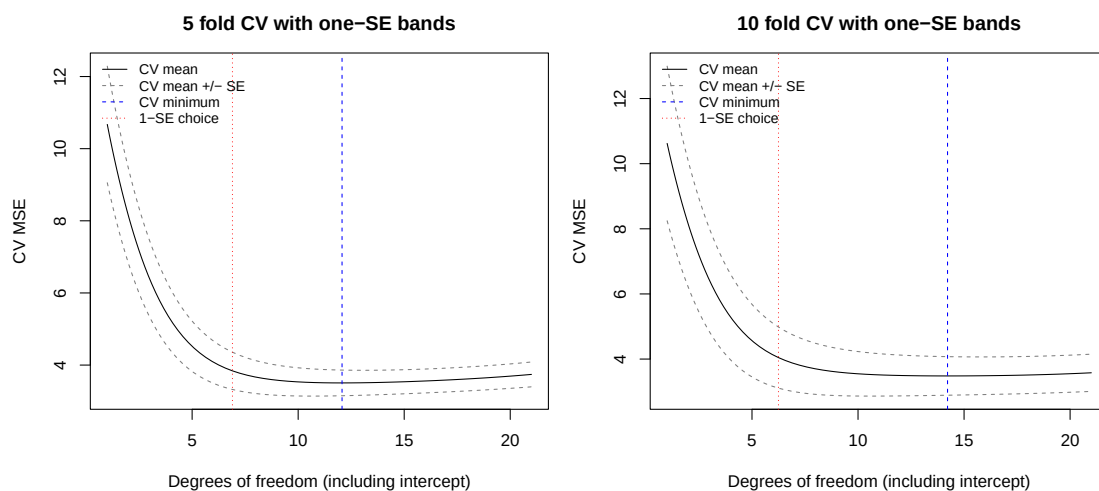
$$CV_K(\lambda) \leq CV_K(\lambda_{\min,K}) + SE_K(\lambda_{\min,K}).$$

Thus the one-SE rule favors stronger regularization among fits whose CV error is sufficiently close to the minimum.

This standard error is a fold-based selection heuristic; overlapping training sets do not produce independent fits.



Ridge training, test, and CV errors. The plots show the full error range and a detailed view.



Ridge CV means with one-SE bands and selected penalties.

Selection	λ	Test MSE
Test-grid oracle	0.132792	4.399062
5-fold CV minimum	0.236711	4.420446
5-fold CV 1-SE	1.063994	4.862883
10-fold CV minimum	0.132792	4.399062
10-fold CV 1-SE	1.340780	5.029434

The test-grid oracle is the smallest observed test error among the 240 fitted penalties. It is reported only as a benchmark: test data are not used to select the CV penalties. A minimum over a finite grid need not equal the exact minimum over all $\lambda \geq 0$.

(c) **Lasso.** The objective is

$$\frac{1}{2n} \|Y - \beta_0 \mathbf{1} - Xb\|_2^2 + \lambda \|b\|_1.$$

The implementation uses `glmnet` with these settings:

```
1 alpha = 1
2 standardize = FALSE
3 intercept = TRUE
```

The script checks the original-scale KKT conditions, including the zero-mean residual condition for the intercept. For Gaussian Lasso, the package's internal response transformation is undone and does not change this objective. The script itself does not standardize the response. Ridge is implemented separately to preserve its stated original-scale tuning convention.

Define

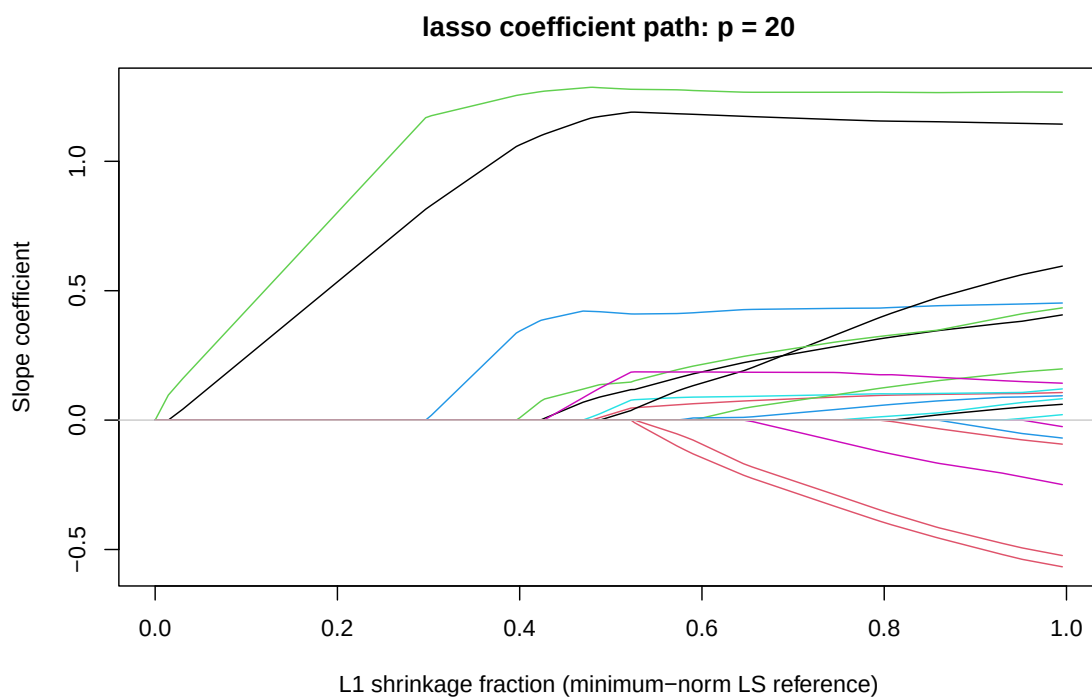
$$\lambda_{\max} = \max_j \left| \frac{1}{n} \tilde{X}_j^\top \tilde{Y} \right|.$$

At this penalty, zero slopes satisfy the stationary conditions. The Lasso grid has 240 log-spaced penalties decreasing from $\lambda_{\max}$ to $10^{-6} \lambda_{\max}$.

The shrinkage fraction is

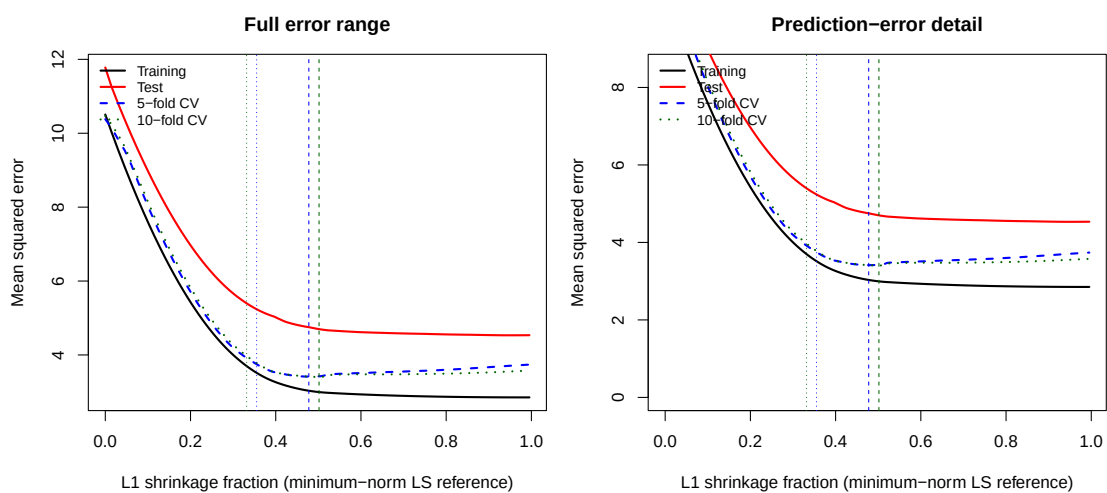
$$f(\lambda) = \frac{\|\hat{b}_\lambda\|_1}{\|\hat{b}_{\text{LS}}\|_1}.$$

For $p = 20$, the centered design has full column rank almost surely, so the least-squares slope reference is unique.

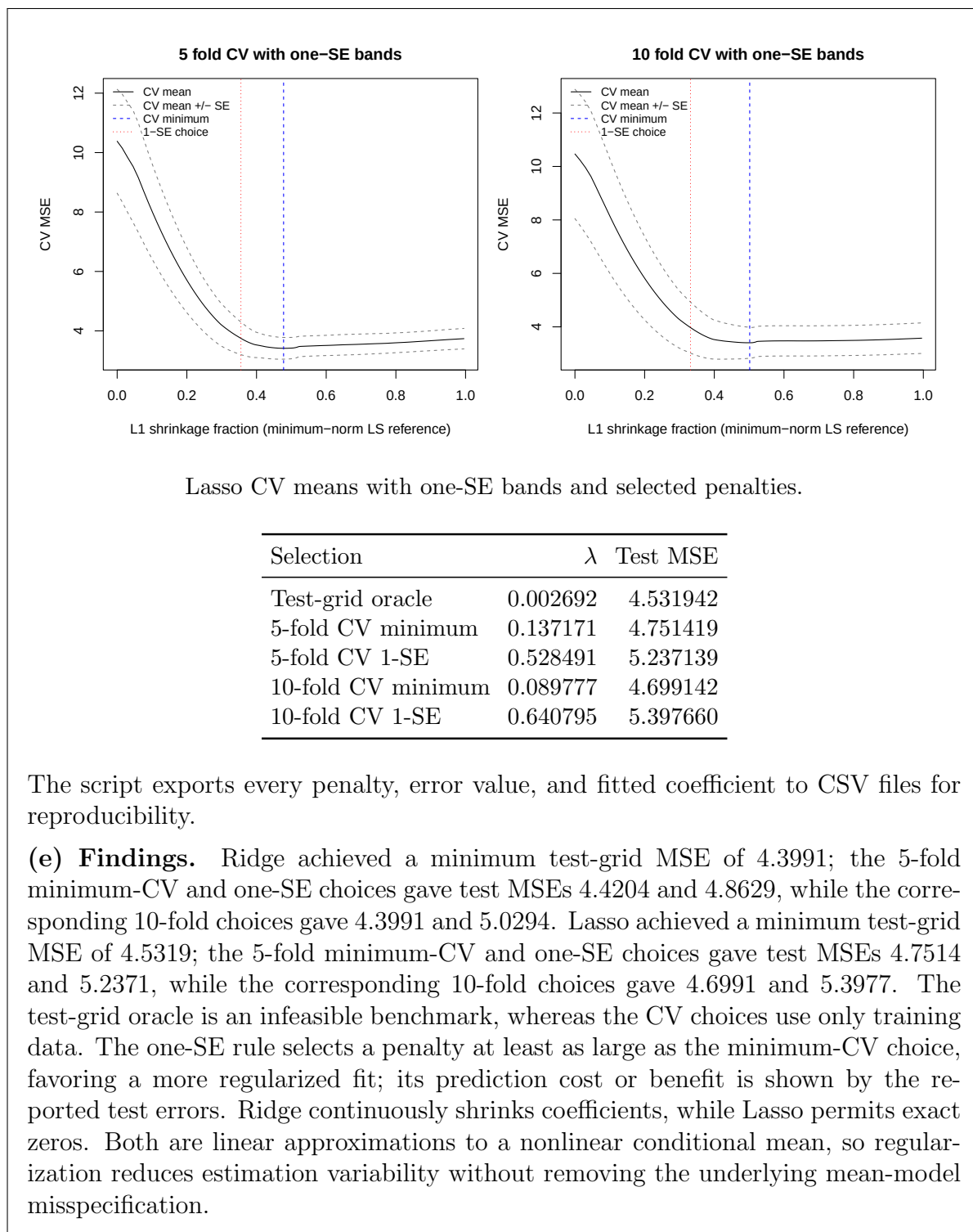


Lasso coefficient path versus the L_1 shrinkage fraction.

(d) Lasso errors. Use the same MSE definitions, folds, and selection rules as in part (b), now plotting against $f(\lambda)$.



Lasso training, test, and CV errors.



6. Exercise 1.18.

Repeat Exercise 1.17 with $p = 100$ and $p = 200$. For each case, answer questions (i)–(iv) of Exercise 1.17, then summarize the findings in one paragraph.

Solution:

Each dimension uses the same simulation code as Exercise 1.17, with only p changed. The seed is reset to zero before each complete training/test simulation, and the CV seed is applied afterward. The fitting methods, error definitions, and selection rules are unchanged.

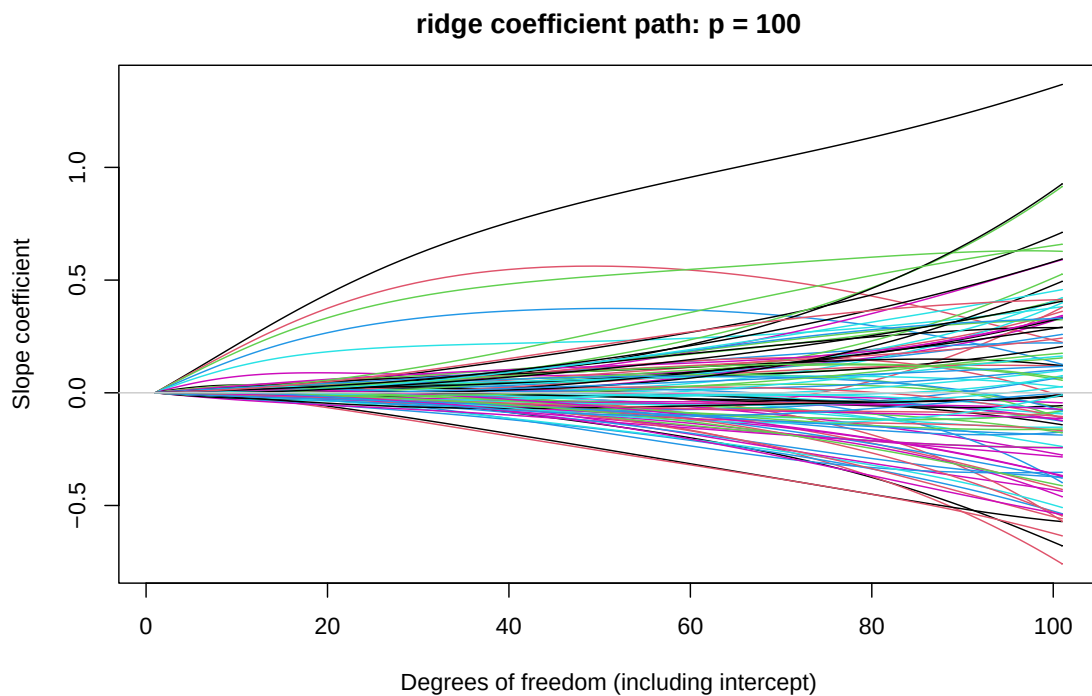
The $p = 100$ case.

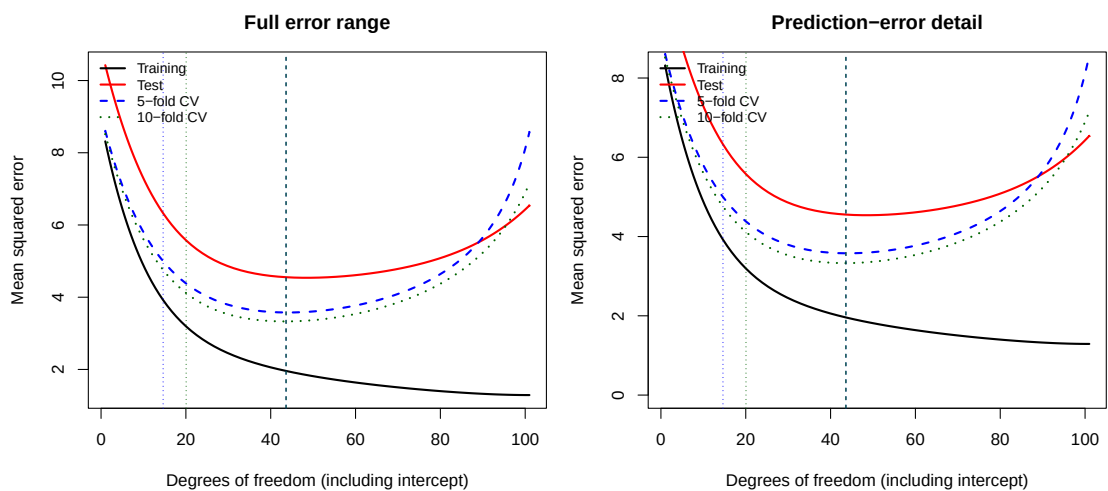
The first training and test rows are

Row	y	x_1	x_2	x_3	x_4	x_5
1	0.105765	0.285543	-0.177280	0.251761	0.429468	0.152709
2	9.928523	1.242860	1.887253	1.807706	1.390074	1.028198
3	-0.864101	-0.727867	-1.229697	-1.478201	0.167907	-0.366386

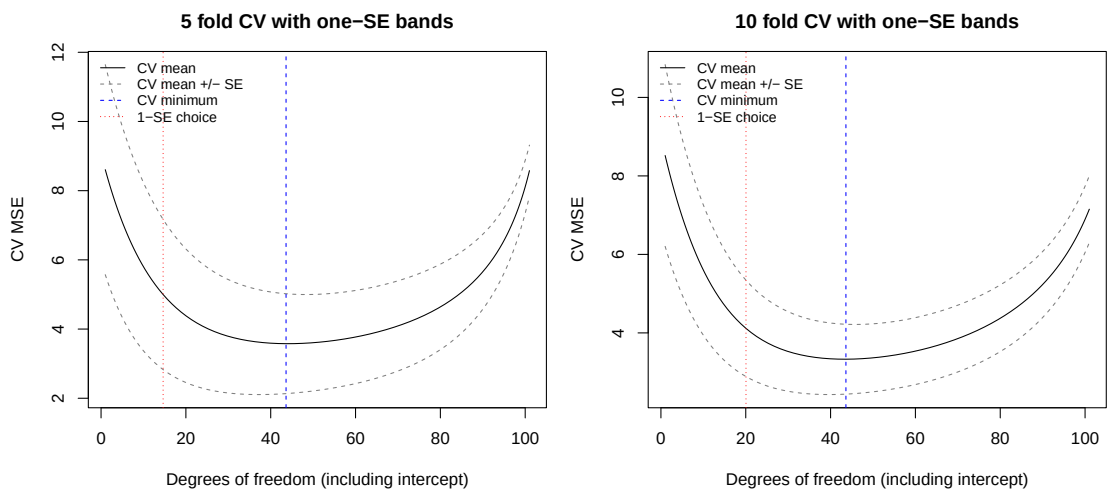
Row	y	x_1	x_2	x_3	x_4	x_5
1	3.672648	1.728780	1.565309	0.905391	-0.117927	-1.381429
2	-0.091899	-0.118988	-0.613030	-1.594482	-0.206495	0.374312
3	-0.443507	0.536662	-0.090179	0.127162	0.618102	1.071354

The ridge results are





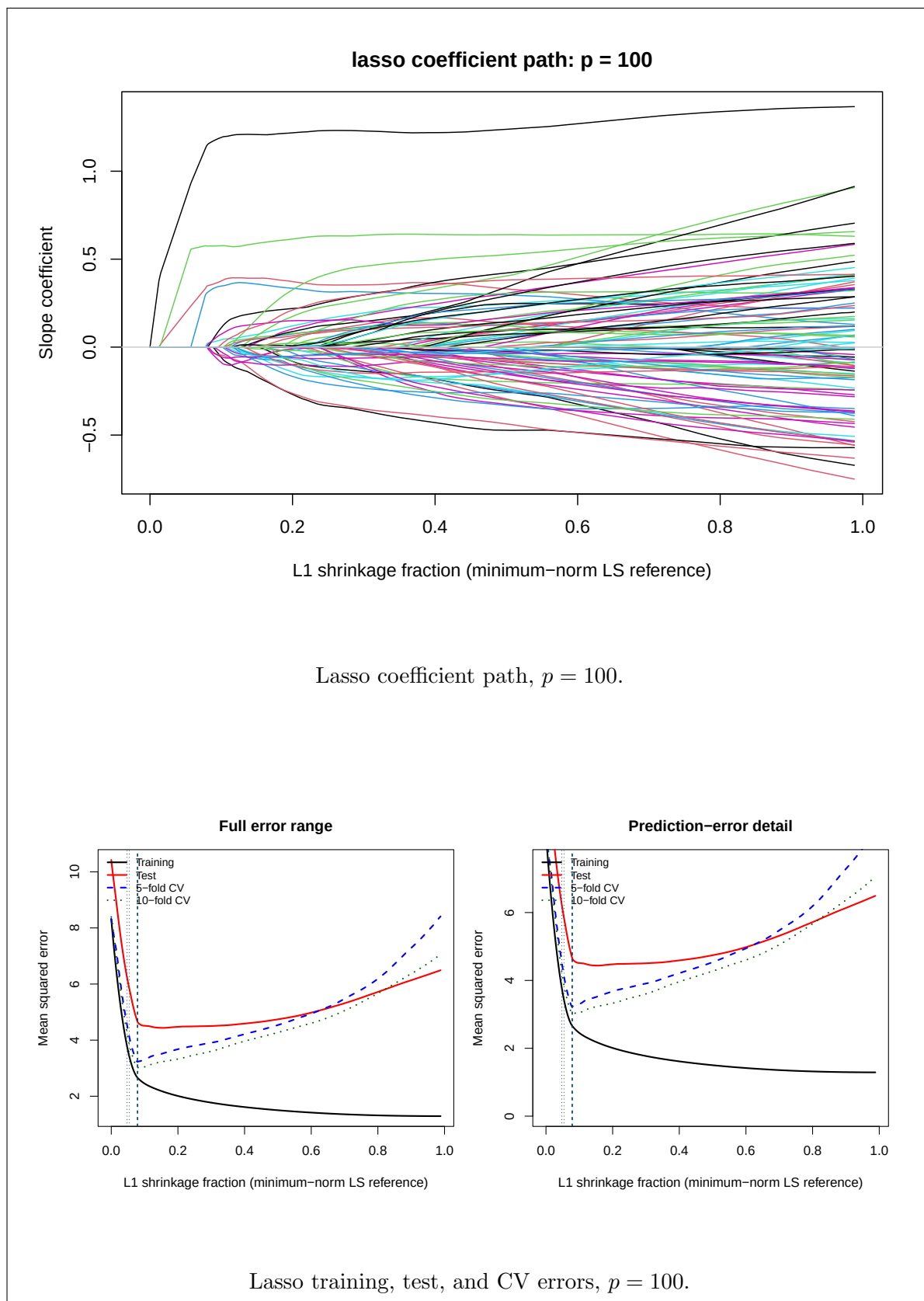
Ridge training, test, and CV errors, $p = 100$.

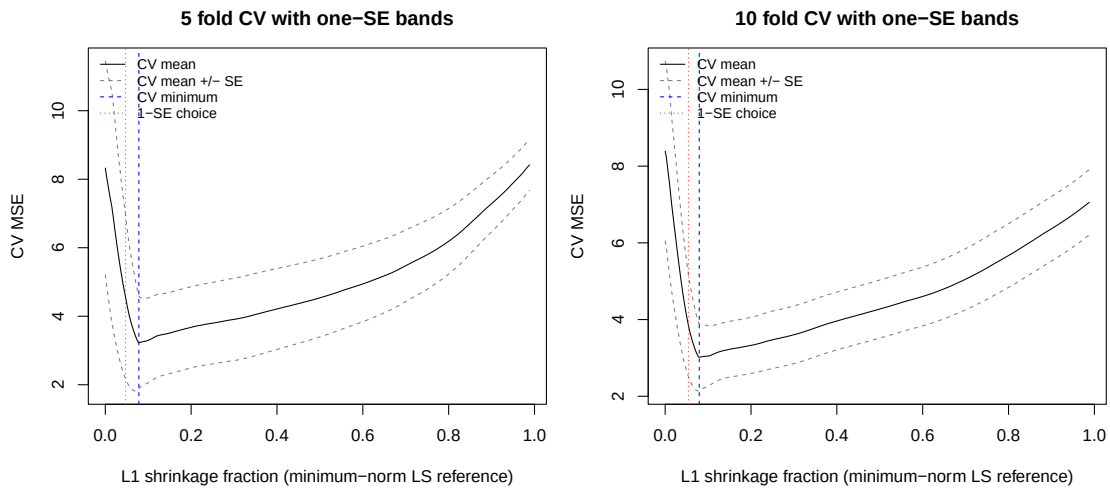


Ridge CV bands and selected penalties, $p = 100$.

Selection	λ	Test MSE
Test-grid oracle	0.262328	4.538391
5-fold CV minimum	0.371085	4.553833
5-fold CV 1-SE	3.746770	6.326653
10-fold CV minimum	0.371085	4.553833
10-fold CV 1-SE	2.101895	5.574011

The Lasso results are





Lasso CV bands and selected penalties, $p = 100$.

Selection	λ	Test MSE
Test-grid oracle	0.068169	4.437006
5-fold CV minimum	0.243156	4.675298
5-fold CV 1-SE	0.973635	6.227192
10-fold CV minimum	0.216609	4.637334
10-fold CV 1-SE	0.802997	5.834924

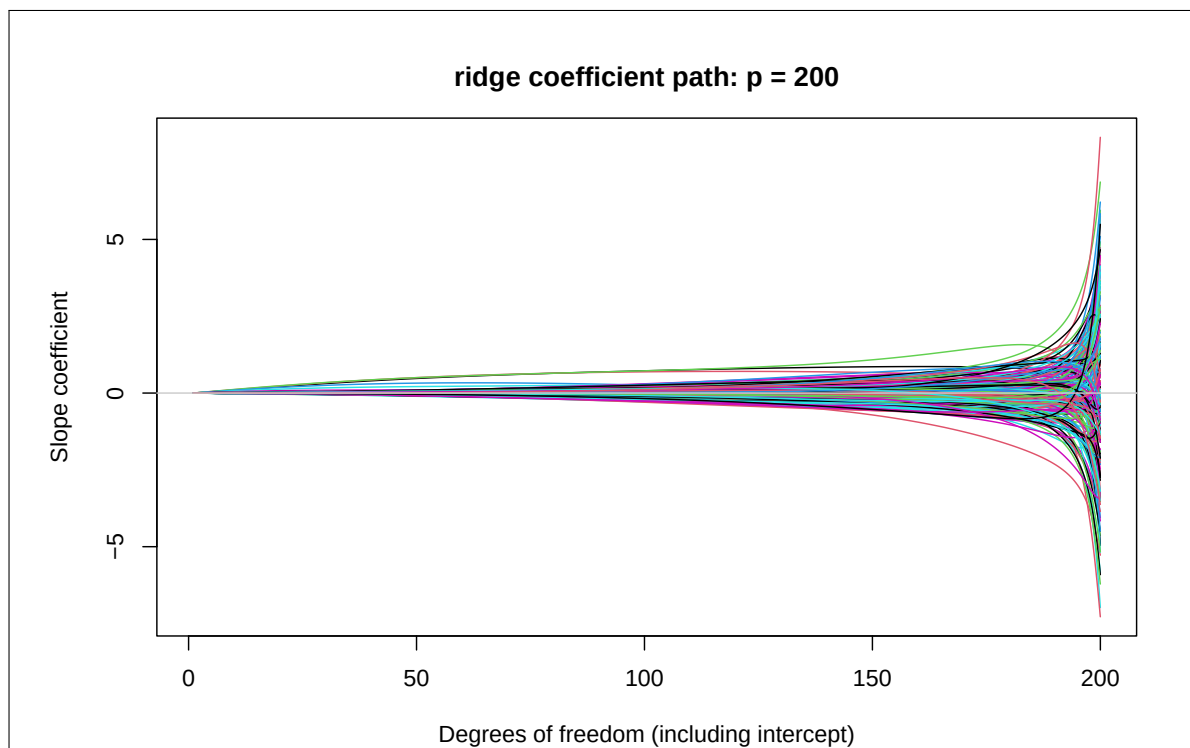
The $p = 200$ case.

The first training and test rows are

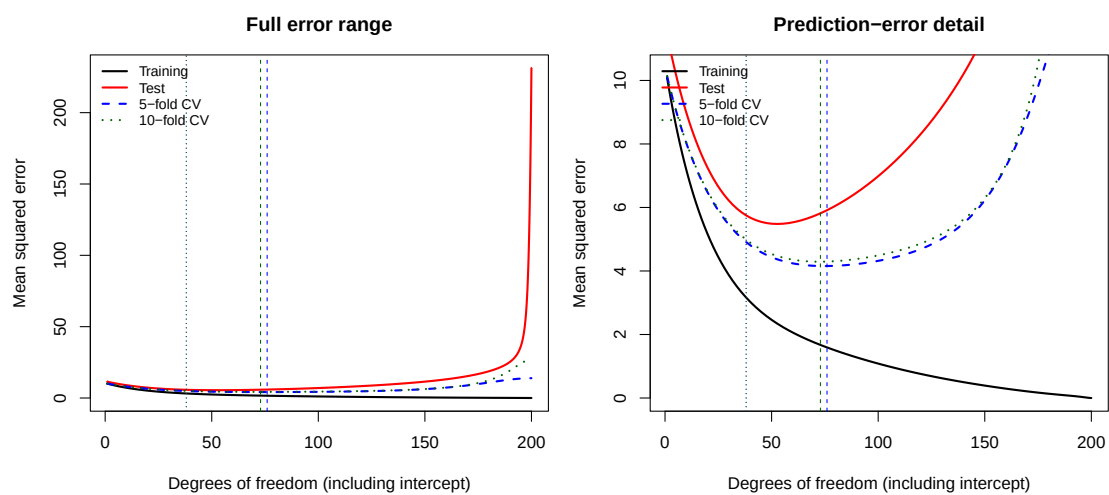
Row	y	x_1	x_2	x_3	x_4	x_5
1	-0.333502	-0.389989	-0.783469	-0.003082	-1.207751	-2.047691
2	-1.084932	-0.826964	-1.294305	-1.778092	-2.587713	-2.136639
3	6.294432	1.185860	1.099989	0.848777	1.817092	2.055061

Row	y	x_1	x_2	x_3	x_4	x_5
1	-0.550661	-0.713210	0.445966	-0.510235	-0.161905	1.572716
2	-0.238505	-0.021047	-0.221596	-0.498722	-0.683671	-0.561626
3	-2.148214	-1.764234	-1.383376	-1.375164	-0.616631	-1.595000

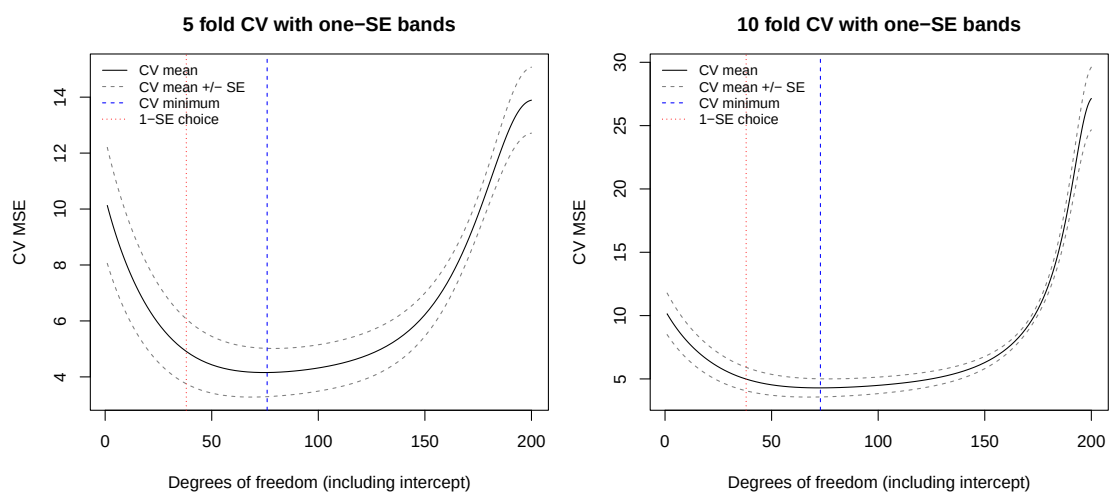
The ridge results are



Ridge coefficient path, $p = 200$.



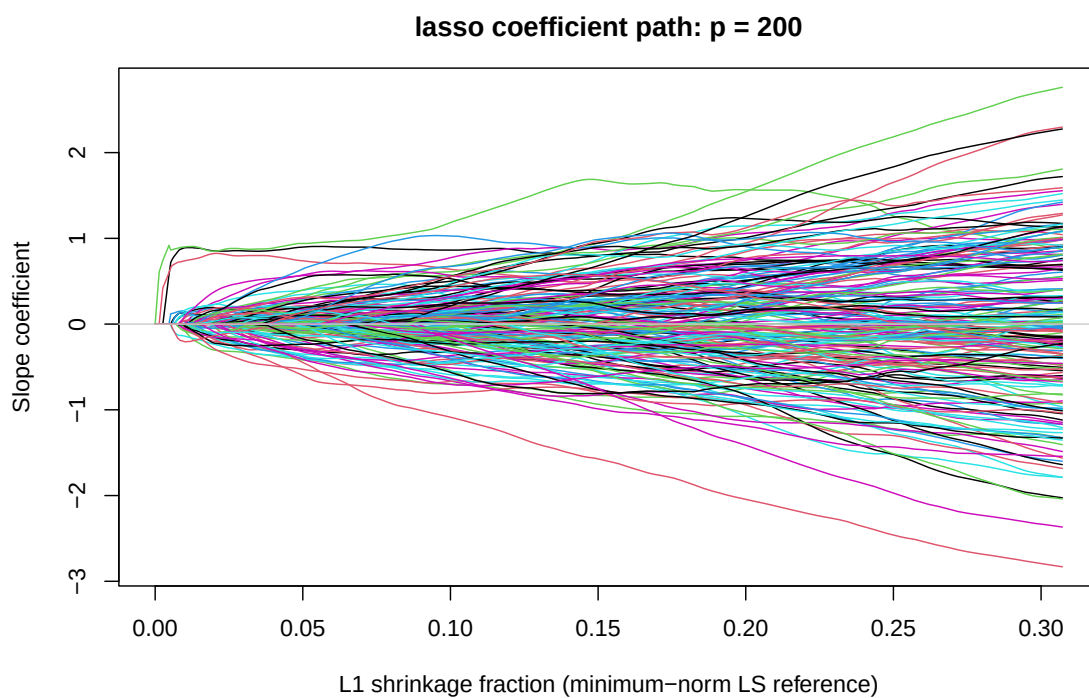
Ridge training, test, and CV errors, $p = 200$.



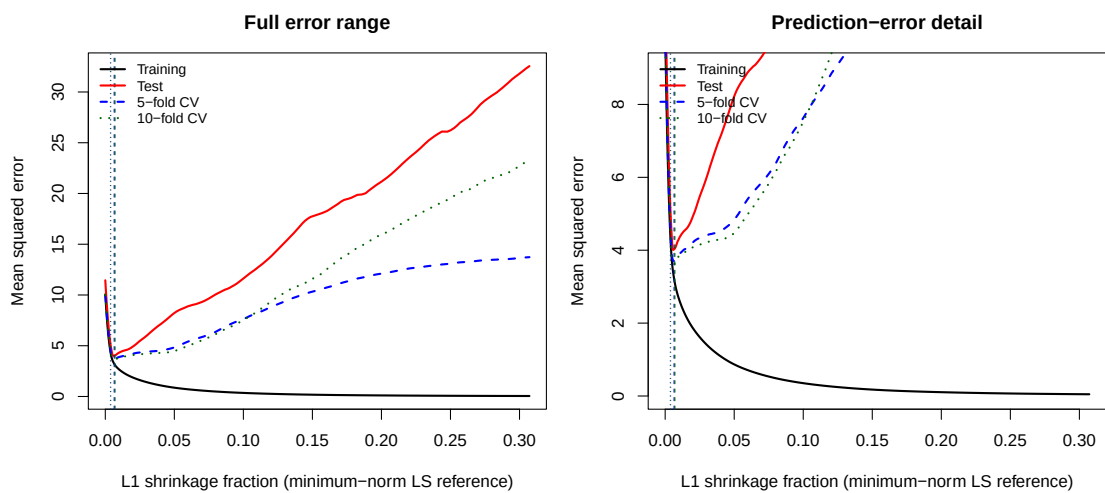
Ridge CV bands and selected penalties, $p = 200$.

Selection	λ	Test MSE
Test-grid oracle	0.993341	5.481090
5-fold CV minimum	0.393935	5.913261
5-fold CV 1-SE	1.987712	5.751401
10-fold CV minimum	0.442215	5.815887
10-fold CV 1-SE	1.987712	5.751401

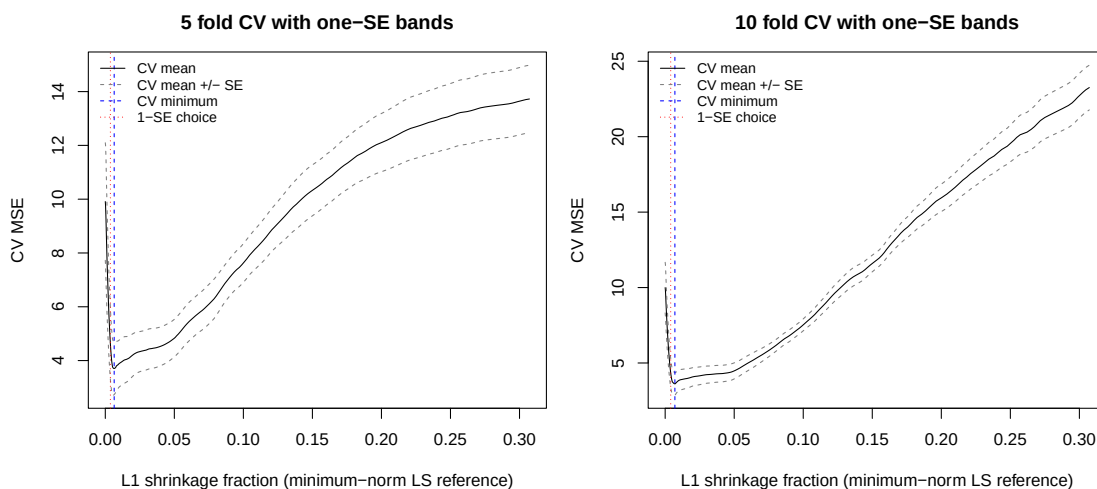
The Lasso results are



Lasso coefficient path, $p = 200$.



Lasso training, test, and CV errors, $p = 200$.



Lasso CV bands and selected penalties, $p = 200$.

Selection	λ	Test MSE
Test-grid oracle	0.305095	4.009939
5-fold CV minimum	0.261511	4.023765
5-fold CV 1-SE	0.932806	5.240093
10-fold CV minimum	0.224153	4.049329
10-fold CV 1-SE	0.799550	4.905401

Rank and reference convention. At $p = 200$, centering $n = 200$ observations gives

$$\text{rank}(\tilde{X}) \leq 199.$$

The unpenalized least-squares slope solution is therefore nonunique. The script uses the minimum-Euclidean-norm solution

$$\hat{b}_{\text{LS}} = \tilde{X}^+ \tilde{Y}$$

as the explicit reference for the Lasso shrinkage fraction.

Different least-squares slope solutions can have different L_1 norms, so this convention matters. With this reference, a shrinkage fraction above 1 is possible; it is not automatically a coding error. The plots preserve penalty-path order rather than assuming the fraction is monotone.

Positive ridge penalties remain well defined even within CV folds whose training sample size is smaller than the number of covariates.

Changing p changes the realized observations, even when the seed is the same. Thus these comparisons are not paired comparisons on identical responses, although the generating mean formula remains unchanged.

Findings. For $p = 100$, Ridge achieved a minimum test-grid MSE of 4.5384; the 5-fold minimum-CV and one-SE choices gave test MSEs 4.5538 and 6.3267, while the corresponding 10-fold choices gave 4.5538 and 5.5740. Lasso achieved a minimum test-grid MSE of 4.4370; the 5-fold minimum-CV and one-SE choices gave test MSEs 4.6753 and 6.2272, while the corresponding 10-fold choices gave 4.6373 and 5.8349. For $p = 200$, Ridge achieved a minimum test-grid MSE of 5.4811; the 5-fold minimum-CV and one-SE choices gave test MSEs 5.9133 and 5.7514, while the corresponding 10-fold choices gave 5.8159 and 5.7514. Lasso achieved a minimum test-grid MSE of 4.0099; the 5-fold minimum-CV and one-SE choices gave test MSEs 4.0238 and 5.2401, while the corresponding 10-fold choices gave 4.0493 and 4.9054. Increasing the dimension adds correlated predictors without changing the four variables in the generating mean. The comparisons reflect both increased estimation complexity and the realized samples. At $p = 200$, the centered design has rank at most 199, making the unpenalized slope solution nonunique, while positive ridge penalties remain well defined. The Lasso fraction uses the minimum-norm least-squares reference. Changing the dimension also changes the random-number sequence, so these are not paired comparisons on identical observations.

Reproducible R code

Place `homework2.R` in the same folder as this document. Run the following commands from the R console:

```
1 setwd("C:/Users/prana/OneDrive/Desktop/MathNotes/Stats 596")
2
3 if (!requireNamespace("MASS", quietly = TRUE)) {
4   install.packages("MASS")
5 }
6
7 if (!requireNamespace("glmnet", quietly = TRUE)) {
8   install.packages("glmnet")
9 }
10
11 source("homework2.R")
```

The script creates the folder `hw2_results`, writes PDF figures, CSV tables, and LaTeX result fragments, and records the R session information. After it finishes, recompile this document.

Inspect any warnings about minima occurring at penalty-grid endpoints before treating those minima as adequately resolved. If numerical output has not yet been generated, this document shows explicit placeholders instead of fabricated results.

```
1 # Stats 596 -- Homework 2 -- Pranav Tikkawar
2 # Save this file as homework2.R beside hw2.tex.
3 # Set the R working directory to that folder before sourcing this script.
4 # Outputs are written to hw2_results/; all names match hw2.tex.
5 # Requires MASS and glmnet. No numerical results are hard-coded.
6
7 options(stringsAsFactors = FALSE, width = 120)
8 needed <- c("MASS", "glmnet")
9 missing <- needed[!vapply(needed, requireNamespace, logical(1), quietly = TRUE)]
10 if (length(missing)) {
11   stop("Install required packages first: install.packages(c(",
12     paste(sprintf('"%s"', missing), collapse = ", "), ")")")
13 }
14 resdir <- "hw2_results"
15 dir.create(resdir, recursive = TRUE, showWarnings = FALSE)
16
17 # LaTeX table exporter. Input headings contain intentional math markup.
18 write.tex.table <- function(d, filename, digits = 6) {
19   text <- lapply(d, function(v) {
20     if (is.numeric(v)) formatC(v, digits = digits, format = "f") else as.
21       character(v)
22   })
23   text <- as.data.frame(text, check.names = FALSE)
24   rows <- apply(text, 1, paste, collapse = " & ")
```

```

24 bs <- intToUtf8(92)
25 cmd <- function(s) paste0(bs, s)
26 ending <- paste0(" ", bs, bs)
27 lines <- c(cmd("begin{center}"), cmd("small"),
28   cmd("setlength{\\tabcolsep}{4pt}"),
29   cmd(paste0("begin{tabular}{l", paste(rep("r", ncol(d)-1), collapse=""), "}"))
30   ,
31   cmd("toprule"), paste0(paste(names(d), collapse=" & "), ending),
32   cmd("midrule"), paste0(rows, ending), cmd("bottomrule"),
33   cmd("end{tabular}"), cmd("end{center}"))
34 writeLines(lines, filename, useBytes = TRUE)
35 }
36
37 # Exercise 1.11: exact normal-mean risks and independent integration checks.
38 ridge.mse <- function(mu, lambda) (1 + lambda^2*mu^2)/(1 + lambda)^2
39 lasso.mse <- function(mu, lambda) {
40   a <- -lambda-mu
41   b <- lambda-mu
42   (1+lambda^2)*(pnorm(a)+pnorm(b, lower.tail=FALSE)) -
43   (a+2*lambda)*dnorm(a) + (b-2*lambda)*dnorm(b) +
44   mu^2*(pnorm(b)-pnorm(a))
45 }
46 checks <- expand.grid(mu=c(-3,0,2), lambda=c(.1,1,5,10))
47 checks$absolute.error <- mapply(function(mu, lambda) {
48   a <- -lambda-mu; b <- lambda-mu
49   value <- integrate(function(u) (u+lambda)^2*dnorm(u), -Inf, a)$value +
50     mu^2*(pnorm(b)-pnorm(a)) +
51     integrate(function(u) (u-lambda)^2*dnorm(u), b, Inf)$value
52   abs(value-lasso.mse(mu,lambda))
53 }, checks$mu, checks$lambda)
54 stopifnot(max(checks$absolute.error)<1e-6)
55 write.csv(checks, file.path(resdir,"ex111_risk_checks.csv"), row.names=FALSE)
56 mu.grid <- seq(-15,15,length.out=1201)
57 lambdas <- c(.1,1,5,10)
58 colors <- c("black","blue","red","darkgreen")
59 pdf(file.path(resdir,"ex111_mse.pdf"), width=11, height=5)
60 par(mfrow=c(1,2), mar=c(4,4,3,1))
61 for (method in c("Ridge","Lasso")) {
62   risk <- if (method=="Ridge") ridge.mse else lasso.mse
63   values <- sapply(lambdas, function(lam) risk(mu.grid,lam))
64   matplot(mu.grid,values,type="l",lty=1,col=colors,lwd=2,
65     xlab=expression(mu),ylab="Mean squared error",main=method)
66   abline(h=1,lty=2,col="gray50")
67   legend("top",paste("lambda =",lambdas),col=colors,lty=1,lwd=2,bty="n")
68   tab <- data.frame(mu=mu.grid,values)
69   names(tab) <- c("mu",paste0("lambda_",lambdas))

```

```

69   write.csv(tab,file.path(resdir,paste0("ex111_",tolower(method),"_mse.csv")),row
    .names=FALSE)
70 }
71 dev.off()
72
73 # Textbook simulation. Generate all data before drawing any CV folds.
74 make.data <- function(p) {
75   set.seed(0)
76   n <- 200
77   mu <- rep(0,p)
78   msig <- toeplitz(.8^(1:p-1))
79   sig <- .5
80   x <- MASS::mvrnorm(n,mu,msig)
81   y <- x[,1] + pmax(0,apply(x[,2:4],1,sum))^2/3 + rnorm(n,0,sig)
82   ntest <- 100
83   xtest <- MASS::mvrnorm(ntest,mu,msig)
84   ytest <- xtest[,1] + pmax(0,apply(xtest[,2:4],1,sum))^2/3 + rnorm(ntest,0,sig)
85   list(x=x,y=y,xtest=xtest,ytest=ytest)
86 }
87
88 # Ridge minimizes  $RSS/(2*n) + \lambda * \text{sum}(\text{slopes}^2)/2$ .
89 # Only centering is performed, to profile out an unpenalized intercept.
90 ridge.path <- function(x,y,lambda) {
91   n <- nrow(x)
92   xb <- colMeans(x); yb <- mean(y)
93   xc <- sweep(x,2,xb,"-"); yc <- y-yb
94   s <- svd(xc)
95   keep <- s$d > max(s$d)*1e-10
96   d <- s$d[keep]
97   if (!length(d)) stop("Centered design has zero rank.")
98   U <- s$u[,keep,drop=FALSE]; V <- s$v[,keep,drop=FALSE]
99   uy <- drop(crossprod(U,yc))
100  W <- vapply(lambda,function(lam) d/(d^2+n*lam),numeric(length(d)))
101  W <- matrix(W,nrow=length(d),ncol=length(lambda))
102  B <- V %*% sweep(W,1,uy,"*")
103  shrink <- vapply(lambda,function(lam) d^2/(d^2+n*lam),numeric(length(d)))
104  shrink <- matrix(shrink,nrow=length(d),ncol=length(lambda))
105  list(beta=B,intercept=yb-drop(xb %*% B),df=1+colSums(shrink))
106 }
107
108 # Gaussian Lasso on original scales with an unpenalized intercept.
109 # alpha=1 preserves the stated Lasso objective when glmnet undoes its
110 # internal response transformation. No manual x or y rescaling is used.
111 lasso.path <- function(x, y, lambda) {
112   fit <- glmnet::glmnet(
113     x = x,

```

```

114     y = y,
115     family = "gaussian",
116     alpha = 1,
117     lambda = lambda,
118     standardize = FALSE,
119     intercept = TRUE,
120     thresh = 1e-10,
121     maxit = 3000000
122 )
123
124 if (length(fit$lambda) != length(lambda)) {
125     stop(
126         "glmnet returned ", length(fit$lambda),
127         " of ", length(lambda), " requested penalties. ",
128         "The path did not finish; do not use incomplete results."
129     )
130 }
131
132 if (max(abs(fit$lambda - lambda)) >
133     1e-10 * max(1, max(lambda))) {
134     stop("Returned penalties do not match the requested grid.")
135 }
136
137 list(
138     beta = as.matrix(fit$beta),
139     intercept = as.numeric(fit$a0),
140     df = NULL
141 )
142 }
143 fit.path <- function(x,y,lambda,method) {
144     if (method=="ridge") ridge.path(x,y,lambda) else lasso.path(x,y,lambda)
145 }
146 predict.path <- function(fit,x) sweep(x %*% fit$beta,2,fit$intercept,"+")
147 path.mse <- function(y,pred) colMeans(sweep(pred,1,y,"-")^2)
148
149 # Equal-size fold errors, fold-based SE, and the largest acceptable lambda.
150 cv.path <- function(x,y,lambda,method,folds) {
151     K <- max(folds)
152     error <- matrix(NA_real_,K,length(lambda))
153     for (k in seq_len(K)) {
154         validation <- which(folds==k)
155         training <- which(folds!=k)
156         fit <- fit.path(x[training,,drop=FALSE],y[training],lambda,method)
157         error[k,] <- path.mse(y[validation],predict.path(fit,x[validation,,drop=FALSE]
158     ))
159 }

```

```

159 average <- colMeans(error)
160 se <- apply(error,2,sd)/sqrt(K)
161 imin <- which.min(average)
162 eligible <- which(average<=average[imin]+se[imin])
163 ilse <- eligible[which.max(lambda[eligible])]
164 list(mean=average,se=se,imin=imin,ilse=ilse,fold.errors=error)
165 }
166
167 # Original-scale Lasso KKT checks, including the intercept equation.
168 lasso.kkt <- function(x,y,fit,lambda) {
169   residual <- sweep(predict.path(fit,x),1,y,"-")
170   gradient <- crossprod(x,residual)/nrow(x)
171   vapply(seq_along(lambda),function(k) {
172     b <- fit$beta[,k]
173     active <- abs(b)>1e-8
174     violations <- numeric(length(b))
175     violations[active] <- abs(gradient[active,k]+lambda[k]*sign(b[active]))
176     violations[!active] <- pmax(abs(gradient[!active,k])-lambda[k],0)
177     max(c(violations,abs(mean(residual[,k]))))
178   },numeric(1))
179 }
180
181 run.method <- function(dat,method,folds5,folds10,p) {
182   x <- dat$x; y <- dat$y
183   xc <- sweep(x,2,colMeans(x),"-"); yc <- y-mean(y)
184   if (method=="ridge") {
185     scale <- max(svd(xc,nu=0,nv=0)$d)^2/nrow(x)
186     lambda <- scale*10^seq(4,-8,length.out=240)
187   } else {
188     lambda.max <- max(abs(drop(crossprod(xc,yc))))/nrow(x)
189     lambda <- lambda.max * 10^seq(
190   from = 0,
191   to = -4,
192   length.out = 240
193 )
194   }
195   fit <- fit.path(x,y,lambda,method)
196   train <- path.mse(y,predict.path(fit,x))
197   test <- path.mse(dat$ytest,predict.path(fit,dat$xtest))
198   cv5 <- cv.path(x,y,lambda,method,folds5)
199   cv10 <- cv.path(x,y,lambda,method,folds10)
200   stem <- paste0("p",p,"_",method)
201   if (method=="ridge") {
202     axis <- fit$df
203     xlabel <- "Degrees of freedom (including intercept)"
204   } else {

```

```

205 reference <- ridge.path(x,y,0)
206 axis <- colSums(abs(fit$beta))/sum(abs(reference$beta[,1]))
207 xlabel <- "L1 shrinkage fraction (minimum-norm LS reference)"
208 checks <- lasso.kkt(x,y,fit,lambda)
209 write.csv(data.frame(lambda=lambda,kkt.violation=checks),
210           file.path(resdir,paste0(stem,"_kkt.csv")),row.names=FALSE)
211 if (max(checks)>1e-4) {
212   stop(stem,": KKT violation exceeds 1e-4; inspect convergence.")
213 }
214 }
215 best <- which.min(test)
216 selected <- c(best,cv5$imin,cv5$i1se,cv10$imin,cv10$i1se)
217 selection <- c("Test-grid oracle","5-fold CV minimum","5-fold CV 1-SE",
218              "10-fold CV minimum","10-fold CV 1-SE")
219 tab <- data.frame(Selection=selection,lambda=lambda[selected],
220                  'Test MSE'=test[selected],check.names=FALSE)
221 write.csv(tab,file.path(resdir,paste0(stem,"_selection.csv")),row.names=FALSE)
222 latex.tab <- tab
223 names(latex.tab)[2] <- "$\\lambda$"
224 write.tex.table(latex.tab,file.path(resdir,paste0(stem,"_selection.tex")))
225 detail <- data.frame(lambda=lambda,axis=axis,train=train,test=test,
226                     cv5=cv5$mean,se5=cv5$se,cv10=cv10$mean,se10=cv10$se)
227 write.csv(detail,file.path(resdir,paste0(stem,"_path.csv")),row.names=FALSE)
228 coef <- rbind(intercept=fit$intercept,fit$beta)
229 rownames(coef) <- c("Intercept",paste0("x",seq_len(ncol(x))))
230 write.csv(coef,file.path(resdir,paste0(stem,"_coefficients.csv")))
231 for (K in c(5,10)) {
232   cv <- if(K==5) cv5 else cv10
233   write.csv(cv$fold.errors,file.path(resdir,paste0(stem,"_cv",K,"_fold_errors.
234   csv")),row.names=FALSE)
235 }
236 at.endpoint <- c(best,cv5$imin,cv10$imin) %in% c(1,length(lambda))
237 if(any(at.endpoint)) {
238   warning(stem,": a test/CV minimum is at a grid endpoint. Inspect and extend
239   the grid if needed.")
240 }
241 list(method=method,lambda=lambda,fit=fit,axis=axis,xlabel=xlabel,
242       train=train,test=test,cv5=cv5,cv10=cv10,table=tab,best=best,
243       endpoint=at.endpoint)
244 }
245
246 plot.result <- function(z,p) {
247   stem <- paste0("p",p,"_",z$method)
248   # Keep penalty-path order; the L1 reference fraction need not be monotone.
249   pdf(file.path(resdir,paste0(stem,"_path.pdf")),width=8,height=5.2)
250   par(mar=c(4.5,4,3,1))

```

```

249  matplot(z$axis,t(z$fit$beta),type="l",lty=1,
250          xlab=z$xlabel,ylab="Slope coefficient",
251          main=paste(z$method,"coefficient path: p =",p))
252  abline(h=0,col="gray80")
253  dev.off()
254
255  pdf(file.path(resdir,paste0(stem,"_errors.pdf")),width=11,height=5)
256  par(mfrow=c(1,2),mar=c(4.5,4,3,1))
257  E <- cbind(z$train,z$test,z$cv5$mean,z$cv10$mean)
258  for (zoom in c(FALSE,TRUE)) {
259    limits <- range(E)
260    if (zoom) {
261      minimum <- min(c(z$test,z$cv5$mean,z$cv10$mean))
262      limits <- c(0,max(1,2.5*minimum))
263    }
264    matplot(z$axis,E,type="l",lty=c(1,1,2,3),lwd=2,
265            col=c("black","red","blue","darkgreen"),ylim=limits,
266            xlab=z$xlabel,ylab="Mean squared error",
267            main=if(zoom) "Prediction-error detail" else "Full error range")
268    abline(v=z$axis[c(z$cv5$imin,z$cv5$i1se)],col="blue",lty=c(2,3))
269    abline(v=z$axis[c(z$cv10$imin,z$cv10$i1se)],col="darkgreen",lty=c(2,3))
270    legend("topleft",c("Training","Test","5-fold CV","10-fold CV"),
271           col=c("black","red","blue","darkgreen"),lty=c(1,1,2,3),
272           lwd=2,bty="n",cex=.8)
273  }
274  dev.off()
275
276  pdf(file.path(resdir,paste0(stem,"_cv.pdf")),width=11,height=5)
277  par(mfrow=c(1,2),mar=c(4.5,4,3,1))
278  for (K in c(5,10)) {
279    cv <- if(K==5) z$cv5 else z$cv10
280    plot(z$axis,cv$mean,type="l",xlab=z$xlabel,ylab="CV MSE",
281         main=paste(K,"fold CV with one-SE bands"),
282         ylim=range(c(cv$mean-cv$sse,cv$mean+cv$sse)))
283    lines(z$axis,cv$mean+cv$sse,col="gray50",lty=2)
284    lines(z$axis,cv$mean-cv$sse,col="gray50",lty=2)
285    abline(v=z$axis[c(cv$imin,cv$i1se)],col=c("blue","red"),lty=c(2,3))
286    legend("topleft",c("CV mean","CV mean +/- SE","CV minimum","1-SE choice"),
287           col=c("black","gray50","blue","red"),lty=c(1,2,2,3),bty="n",cex=.8)
288  }
289  dev.off()
290 }
291
292 results <- list()
293 for (p in c(20,100,200)) {
294   cat("\n===== p =",p,"=====\n")

```

```

295 dat <- make.data(p)
296 for (kind in c("train","test")) {
297   raw <- if(kind=="train") cbind(dat$y,dat$x) else cbind(dat$ytest,dat$xtest)
298   d <- data.frame(Row=as.character(1:3),raw[1:3,1:6],check.names=FALSE)
299   names(d) <- c("Row","$y$",paste0("$x_",1:5,"$"))
300   cat("\n",kind,"data: first rows\n")
301   print(d)
302   write.tex.table(d,file.path(resdir,paste0("p",p,"_",kind,"_rows.tex")))
303   write.csv(d,file.path(resdir,paste0("p",p,"_",kind,"_rows.csv")),row.names=
FALSE)
304 }
305 set.seed(123)
306 permutation <- sample.int(nrow(dat$x))
307 folds5 <- folds10 <- integer(nrow(dat$x))
308 folds5[permutation] <- rep(1:5,length.out=nrow(dat$x))
309 folds10[permutation] <- rep(1:10,length.out=nrow(dat$x))
310 write.csv(data.frame(row=seq_len(nrow(dat$x)),fold5=folds5,fold10=folds10),
311   file.path(resdir,paste0("p",p,"_folds.csv")),row.names=FALSE)
312 results[[as.character(p)]] <- list()
313 for (method in c("ridge","lasso")) {
314   cat("\nFitting",method,"and both CV paths...\n")
315   z <- run.method(dat,method,folds5,folds10,p)
316   plot.result(z,p)
317   print(z$table,row.names=FALSE)
318   results[[as.character(p)]] [[method]] <- z
319 }
320 }
321
322 # Numerical summaries are generated from the actual computed tables.
323 fmt <- function(x) sprintf("%.4f",x)
324 method.sentence <- function(z) {
325   t <- z$table
326   paste0(if(z$method=="ridge") "Ridge" else "Lasso",
327     " achieved a minimum test-grid MSE of ",fmt(t[[3]][1]),
328     "; the 5-fold minimum-CV and one-SE choices gave test MSEs ",
329     fmt(t[[3]][2])," and ",fmt(t[[3]][3]),
330     ", while the corresponding 10-fold choices gave ",
331     fmt(t[[3]][4])," and ",fmt(t[[3]][5]),". ")
332 }
333 r20 <- results[["20"]]
334 summary17 <- paste0(method.sentence(r20$ridge),method.sentence(r20$lasso),
335   "The test-grid oracle is an infeasible benchmark, whereas the CV choices use
only training data. ",
336   "The one-SE rule selects a penalty at least as large as the minimum-CV choice,
favoring ",
337   "a more regularized fit; its prediction cost or benefit is shown by the

```

```

    reported test errors. ",
338 "Ridge continuously shrinks coefficients, while Lasso permits exact zeros. Both
    are linear ",
339 "approximations to a nonlinear conditional mean, so regularization reduces
    estimation ",
340 "variability without removing the underlying mean-model misspecification.")
341 writeLines(summary17,file.path(resdir,"ex117_summary.tex"))
342 summary18 <- paste0("For $p=100$, ",method.sentence(results[["100"]]$ridge),
343   method.sentence(results[["100"]]$lasso),"For $p=200$, ",
344   method.sentence(results[["200"]]$ridge),method.sentence(results[["200"]]$lasso)
    ,
345 "Increasing the dimension adds correlated predictors without changing the four
    variables ",
346 "in the generating mean. The comparisons reflect both increased estimation
    complexity and ",
347 "the realized samples. At $p=200$, the centered design has rank at most 199,
    making the ",
348 "unpenalized slope solution nonunique, while positive ridge penalties remain
    well defined. ",
349 "The Lasso fraction uses the minimum-norm least-squares reference. Changing the
    dimension ",
350 "also changes the random-number sequence, so these are not paired comparisons
    on identical observations.")
351 writeLines(summary18,file.path(resdir,"ex118_summary.tex"))
352 combined <- do.call(rbind,lapply(names(results),function(p) {
353   do.call(rbind,lapply(c("ridge","lasso"),function(method) {
354     data.frame(p=as.integer(p),Method=method,results[[p]][[method]]$table,check.
      names=FALSE)
355   }))
356 }))
357 write.csv(combined,file.path(resdir,"all_selection_results.csv"),row.names=FALSE)
358 saveRDS(results,file.path(resdir,"all_results.rds"))
359 capture.output(sessionInfo(),file=file.path(resdir,"sessionInfo.txt"))
360
361 # Verify every file imported by the supplied LaTeX document exists.
362 expected <- c("ex111_mse.pdf","ex117_summary.tex","ex118_summary.tex")
363 for (p in c(20,100,200)) {
364   expected <- c(expected,paste0("p",p,"_",c("train","test"),"_rows.tex"))
365   for (method in c("ridge","lasso")) {
366     stem <- paste0("p",p,"_",method)
367     expected <- c(expected,paste0(stem,c("_path.pdf","_errors.pdf","_cv.pdf","_
      selection.tex"))))
368   }
369 }
370 stopifnot(all(file.exists(file.path(resdir,expected))))
371 cat("\nFinished. All",length(expected),"LaTeX input files were generated.\n")

```

```
372 cat("Inspect any grid-endpoint warnings, then compile hw2.tex.\n")
```